

FOREWORD

You are now the proud owner of the TRA CHANG Walk Behind Tractor. This is a product of KUBOTA quality engineering and manufacturing. It is made of fine materials and under a rigid quality control system. The walk behind tractor will give you long, satisfactory service.

To obtain the best use of your walk behind tractor, please read this manual carefully. It will help you become familiar with the operation of the walk behind tractor and contains many helpful hints about the walk behind tractor maintenance. In case of further information, please do not hesitate to consult with KUBOTA distributors and dealers.

KUBOTA may change the specifications and other related information about the products without prior notice.

SAFETY FIRST

This symbol, the industry's "Safety Alert Symbol", is used throughout this manual and on labels on the machine itself to warn of the possibility of personal injury or death. Read these instructions carefully. It is essential that you read the instructions and safety regulations before you attempt to assemble or use this walk behind tractor.

-  **DANGER** : Indicates an imminently hazardous situation which, if not avoided, will result in death.
-  **WARNING** : Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
-  **CAUTION** : Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.
-  **IMPORTANT** : Indicates that equipment or property damage could result if instructions are not followed.
-  **NOTE** : Gives helpful information.

KUBOTA LIMITED WARRANTY

After purchase the walk behind tractor, please fill in the warranty receipt record and send it back in order to obtain warranty periods for 3 years. We will repair or replace any part of TRA CHANG Walk Behind Tractor covered by this warranty which becomes defective, malfunctions in workmanship. This warranty does not cover damage, malfunctions or failures resulting from

1. Negligence of performance of required Operation and Maintenance Manual of walk behind tractor.
2. Damages that occur from incorrect operation and neglected maintenance as specified in Operation and Maintenance Manual.
3. Damages that occur from natural calamities or accident.
4. Damages that occur from improper modification and repairing.
5. Damages that occur from not maintenance by KUBOTA dealer.
6. Damages that occur from using the non-genuine KUBOTA parts, unqualified or low-quality lubricants as specified in the Operation and Maintenance Manual.
7. Damages that occur from worn or rust of exterior parts from vibration or friction.
8. Damages that occur from the noise or vibration which not affect to performance.

The company disclaims any costs originated from inconvenience or loosing of the business opportunity during the machine damaged or repairing period. If this product is exported outside the country or its territories, the warranty is void. Warranty card cannot be reissued if lost.

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SAFE OPERATION

Read and understand this manual carefully before operating the walk behind tractor or any implement attached to it. These important precautions, if followed at all times, will help you operate safely.

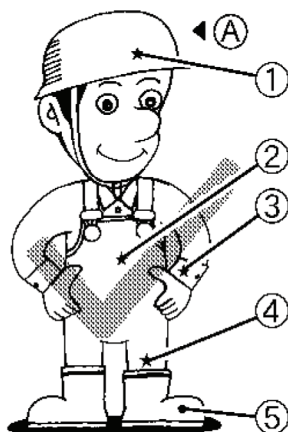
BEFORE OPERATING THE WALK BEHIND TRACTOR

1. Read and understand all instructions and precautions found in both the engine and the walk behind tractor operator's manuals before using. Lack of knowledge can lead to accidents.
2. It is the owner's responsibility to ensure that anyone who will operate the walk behind tractor reads this manual first and becomes familiar with the safe operation.
3. Visually check for abnormal, missing, or malfunctioning parts. Make necessary repairs before operating.
4. Replace damaged or illegible safety labels.
5. Assemble, remove and reinstall the walk behind tractor only as directed in this manual. Failure to do this could result in serious personal injury or death.

OPERATING THE WALK BEHIND TRACTOR

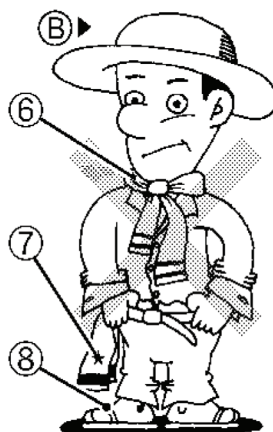
1. Move and turn the walk behind tractor at low speeds.
2. Keep children, others and livestock away when operating the walk behind tractor.
3. For more stability of the walk behind tractor while operating, wheel spacing should be properly adjusted to the work.
4. Please operate the walk behind tractor for the specified work.

5. Be extra careful when operating the walk behind tractor on a slope, always operate up and down, never across the slope. Do not operate on steep slopes or unstable surfaces.
6. When operating the walk behind tractor, do not wear loose or torn clothes because you may get caught by moving parts and wear a safety hat, safety boots or shoes, protective glasses, earplugs or protective gloves, etc.



(A) Good example

- (1) Safety hat
- (2) Overall or protective wear
- (3) Tightly fit sleeves
- (4) Neatly fit pants bottoms
- (5) Anti-slip shoes



(B) Bad example

- (6) Neck towel
- (7) Waist towel
- (8) Sandals

AFTER OPERATING THE WALK BEHIND TRACTOR

1. When finishing operating, park the walk behind tractor on the flat and hard ground. Then lower, stop the engine and lock the brakes.
2. Make sure the detached walk behind tractor stands on a hard, level surface.

SERVICING THE WALK BEHIND TRACTOR

1. Always wear proper protection when servicing or repairing the machine.
2. Do not modify the walk behind tractor. Unauthorized modification may affect the function of the walk behind tractor, which may result in personal injury.
3. The walk behind tractor should not be repaired or maintained while the engine is operating because it may result in personal injury.

CARE OF DANGER, WARNING AND CAUTION LABELS

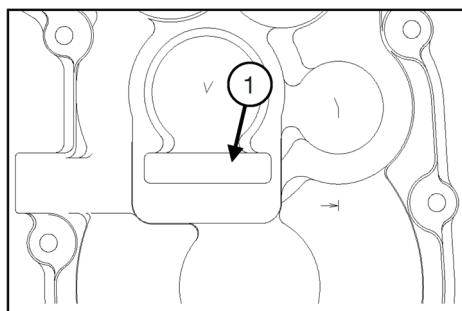
1. Keep danger, warning and caution labels clean and free from obstructing material.
2. Clean danger, warning and caution labels with soap and water, dry with a soft cloth.
3. Replace damaged or missing danger, warning and caution labels with new labels from your local KUBOTA dealer.
4. If a component with danger, warning and caution label (s) affixed is replaced with new part, make sure new label (s) is (are) attached in the same location (s) as the replaced component.
5. Mount new danger, warning and caution labels by applying on a clean dry surface and pressing any bubbles to outside edge.

AFTER SALES SERVICE

Your dealer is interested in your new TRA CHANG walk behind tractor and has the desire to help you get the most value from it.

After reading this manual thoroughly, you will find that you can do the regular maintenance yourself.

However, when in need of parts or major service, be sure to see your local KUBOTA dealer by giving your dealer the walk behind tractor serial number at gear case (See fig.)



(1) Serial number

Please fill in details in the space below.

TRA CHANG Walk Behind Tractor

Model:

Serial number:

Date of purchase:

Name of dealer:

SPECIFICATIONS AND PARTS NAME

SPECIFICATIONS OF THE WALK BEHIND TRACTOR

Walk behind tractor model			NC131	NC131 Special
Model name			NC Plus X	NC Plus X Special
Proper model of KUBOTA diesel engine			8 - 15.5 horsepower	
Brake system			Drum brake type	
Clutch system			Dry multi-plate clutch	
Power engaging and disengaging system			Single clutch	
Gear shift system			Sliding mesh and constant mesh	
Gear position	Gear ratio (between clutch pulley and wheel shaft)	Engine speed (rpm)	Maximum speed of the walk behind tractor tire* (km/h)	
1st gear	44:1	2400	4.2	2.55
2st gear	23:1		8.4	5.1
3st gear	8:1		22.4	13.6
Gear R	56:1		3.4	2.1
Diameter of cage wheel		mm	800	880/980
Size of engine pulley		mm (inch)	140(5.5)	89(4)
Size of the walk behind tractor pulley		mm (inch)	180(7)	
Belt of the walk behind tractor		mm (inch)	1981(78)	1854(73)
Size of hitch-pin hole		mm	26	
Tire*	Type		Tire of the walk behind tractor	
	Dimension		4.00-12	
	Tire pressure		0.32 MPa	
Belt	Name of dimension		B78 V-belt	
	Length	mm	1981	
	Amount	belt	2	
Wheel shaft	Type		Hex shaft	
	Dimension	mm	44.5	
Gear oil		liter	SAE EP90 API GL-4, quantity 5.5 - 6 liters	
Weight (excluding engine)		kg	215	245
Mud guard			-	✓
Large size wheel (880 and 980 mm)			-	✓

Note: The values shown are estimated by manufacturer and are subject of change at any time without notice.

DIMENSIONAL SPECIFICATIONS OF THE WALK BEHIND TRACTOR

The walk behind tractor Model NC Plus X

Model of the walk behind tractor			NC Plus X
B	The width of wheel base from the middle of the wheel	Millimeter	767 - 917
C	The lowest distance from ground	Millimeter	297
D	The height of hitch bar from ground	Millimeter	407
Do	The diameter of the cage wheel	Millimeter	806
E	The distance between the handles	Millimeter	607
Gh	The horizontal distance between the center of gravity and the vertical wheel axle	Millimeter	300
Gv	The vertical distance between the center of gravity and the horizontal wheel axle	Millimeter	260
H	The height of the main clutch lever at its highest position	Millimeter	1147
L	The overall length	Millimeter	2961
W	The width of wheel base from the outer of the wheel	Millimeter	1050 - 1200

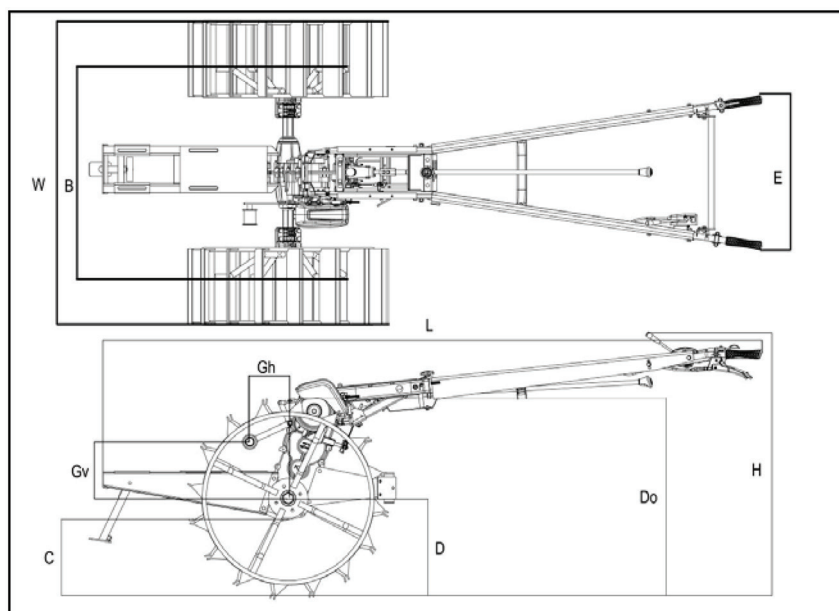
Note: The values shown are estimated by manufacturer and are subject of change at any time without notice.

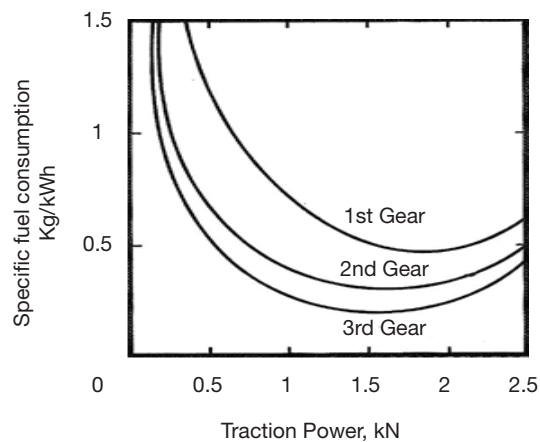
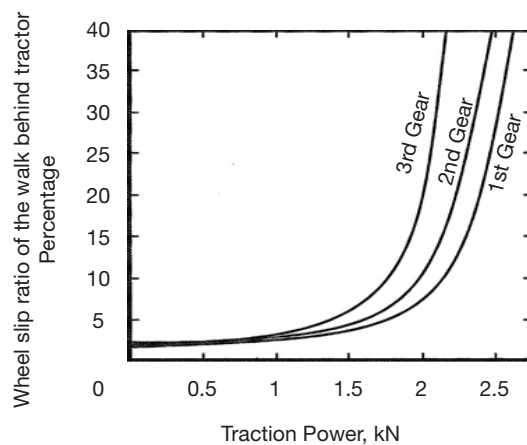
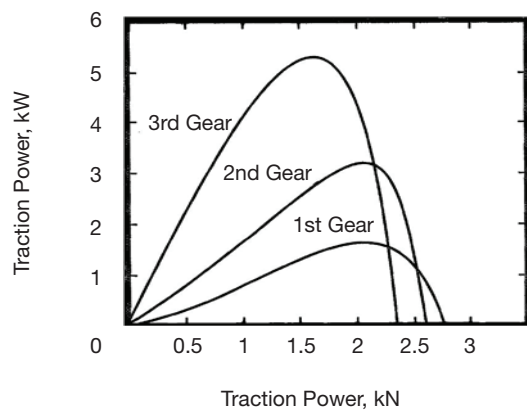
The walk behind tractor

Model NC Plus X Special

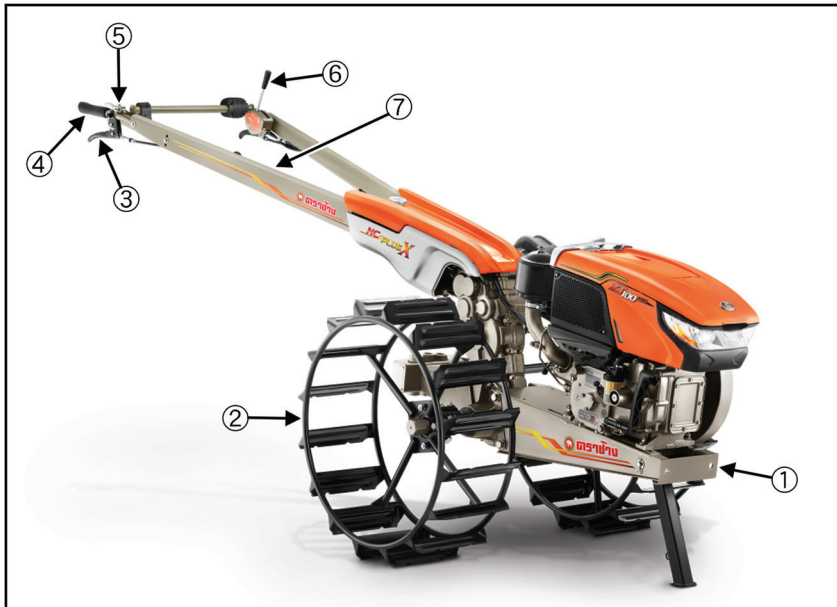
Model of the walk behind tractor			NC Plus X Special
B	The width of wheel base from the middle of the wheel	Millimeter	871 - 1021
C	The lowest distance from ground	Millimeter	357
D	The height of hitch bar from ground	Millimeter	476
Do	The diameter of the cage wheel	Millimeter	938
E	The distance between the handles	Millimeter	607
Gh	The horizontal distance between the center of gravity and the vertical wheel axle	Millimeter	605
Gv	The vertical distance between the center of gravity and the horizontal wheel axle	Millimeter	205
H	The height of the main clutch lever at its highest position	Millimeter	1279
L	The overall length	Millimeter	2961
W	The width of wheel base from the outer of the wheel	Millimeter	1370 - 1270

Note: The values shown are estimated by manufacturer and are subject of change at any time without notice.





PARTS NAME OF THE WALK BEHIND TRACTOR



(1) Stand

(2) Wheel and large size wheel

(3) Steering lever

(4) Handle

(5) Accelerator lever

(6) Main clutch lever (only model
NC Plus X)

(7) Gear shift lever (only model
NC Plus X)



- | | |
|--|--|
| (1) Stand | (7) Gear shift lever or left clutch lever (only model NC Plus X Special) |
| (2) Wheel and large size wheel | (8) Mud guard (only model NC Plus X Special) |
| (3) Steering lever | |
| (4) Handle | |
| (5) Accelerator lever | |
| (6) Right clutch lever or main clutch lever (only model NC Plus X Special) | |

INSTALLING, DISASSEMBLING AND STORING THE WALK BEHIND TRACTOR

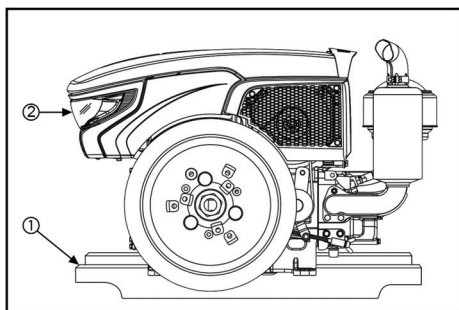
CAUTION

- Due to the large size and heavy weight of the walk behind tractor parts and engine, be extra careful when installing, disassembling and storing and follow the instructions in the manual strictly.

INSTALLING THE ENGINE ON THE WALK BEHIND TRACTOR

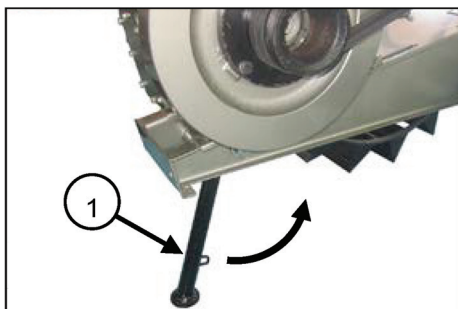
■ IMPORTANT

- KUBOTA diesel engine with an 8-14 horse power can be installed on the TRA CHANG Walk Behind Tractor with no need for wood or iron support.
1. Loosen the nut and remove the engine wood support.



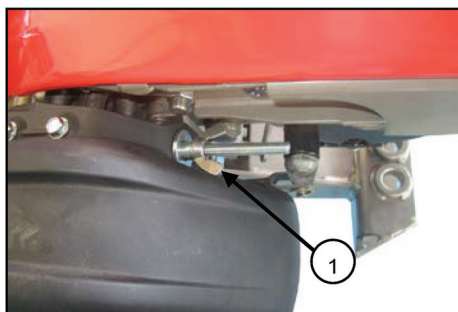
(1) Engine wood support
(2) Engine

2. Set the engine on the walk behind tractor frame with its headlamp turned towards the front of the walk behind tractor. Then tighten the engine securing nuts with the walk behind tractor loosely.
3. Install the pulley on the engine and install the two belts over the engine pulley and main pulley.
4. Fold up the stand at the front of the walk behind tractor, lowering the front to the ground to allow the engine to slide down and tension the belt.



(1) Stand

5. Tighten the four engine securing nuts with the walk behind tractor firmly.
6. Press down using one finger on the middle of the backside of the two belts and tighten the belt tension adjusting wing bolt to tension the belt, following the required belt tension.



(1) Belt tension adjusting wing bolt

Belt tension	30-40 millimeters
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DISASSEMBLING THE ENGINE FROM THE WALK BEHIND TRACTOR

1. Untighten the belt tension adjusting wing bolt to loosen the belt.
2. Untighten the four engine securing nuts loosely.
3. Move the engine to the walk behind tractor to loosen the belt.
4. Remove the two belts from the engine pulley and main pulley.
5. Remove the four engine securing nuts and lift the engine out of the stand of the walk behind tractor.
6. Install the engine on the engine wood support.
7. Storing the walk behind tractor and the engine in the proper place.

STORING THE WALK BEHIND TRACTOR

■ IMPORTANT

- It is very important to store the walk behind tractor properly according to the instructions. This will prolong its service life and prevent rust.

STORING AFTER OPERATION

1. Right after operation, wash mud and soil off the walk behind tractor, especially the wheel axle seal, main clutch mechanism, steering lever and transmission case. Then wipe dry with clean cloth.
2. Lubricate all moving parts with lubricating oil.
3. Store it in the garage.

LONG-TERM STORING

1. Right after operation, wash mud and soil off the walk behind tractor, especially the wheel axle seal, main clutch mechanism, steering lever and transmission case. Then wipe dry with clean cloth.
2. Lubricate all moving parts with lubricating oil.
3. Repair any paint chips at the crashed or struck areas occurred during operation to prevent rusting.
4. Store it in the garage without dust and cover with clean cloth.
5. The main clutch lever should be placed at the **“ENGAGED”** position. If it is placed at the **“STOP”** or the **“DISENGAGED”** position, it may cause a brake lock.

OPERATING THE WALK BEHIND TRACTOR

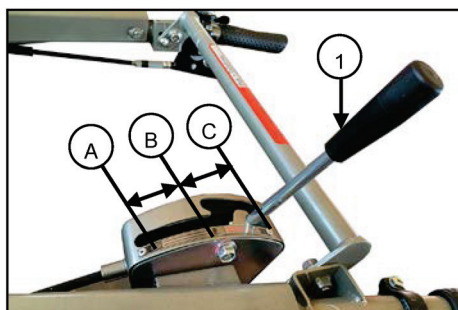
ENGAGING AND DISENGAGING THE ENGINE POWER

The engine power can be engaged and disengaged by using the main clutch lever which has 3 positions:

“ENGAGED” position means the power from the engine is transmitted to the walk behind tractor.

“DISENGAGED” position means the power from the engine is not transmitted to the walk behind tractor.

“STOP” position means the power from the engine is not transmitted to the walk behind tractor. The brake will operate.



- (1) Main clutch lever
- (A) “ENGAGED” position
- (B) “STOP” position
- (C) “DISENGAGED” position

CAUTION

- Slowly engage the main clutch to prevent the walk behind tractor from jerking forward. Otherwise, the engine may stop.
- Do not suddenly pull the main clutch lever to the "STOP" position. Otherwise, the walk behind tractor may come to a sudden stop and you may lose control of the walk behind tractor.

CHANGING THE SPEED

Changing the speed by using the gear shift lever, consisting of three forward gears, one reverse gear and neutral gear as follows:

- | | | |
|-----------------------|-------|--|
| “1st gear” | means | the walk behind tractor runs at low speeds, suitable for plowing, leveling the soil with implements, such as disc plow, mouldboard plow and harrow, and traveling. |
| “2nd gear” | means | the walk behind tractor runs at medium speeds, suitable for towing the trailer. |
| “3rd gear” | means | the walk behind tractor runs at high speeds, suitable for towing the trailer. |
| “Gear R” | means | the walk behind tractor runs backward, suitable for reversing the walk behind tractor towing the trailer. |
| “Gear NEUTRAL” | means | the engine is operating but the walk behind tractor does not move. |



(1) Gear shift lever

Stop the walk behind tractor whenever you want to change the gear. While the engine is operating, if the gear lever cannot be placed as desired, place the gear lever to the **“NEUTRAL”** position and slightly place the main clutch lever to the **“ENGAGED”** position. Then place it back to its position and place the gear lever to the desired position again. You can change the gear easily this time.

CAUTION

- Do not shift gears if the walk behind tractor has not come to a complete stop. Otherwise, the gear teeth and other parts may be damaged.
- Before shifting gears, always pull the main clutch lever to the “STOP” position first.
- When driving the walk behind tractor on a slippery or narrow road, use the “1st” gear.
- When driving up - down a steep slope, do not disengage the main clutch or shift gears to the “NEUTRAL” position. Otherwise, accidents may occur.
- When hitching with a trailer, make sure the trailer is equipped with a brake system.

STOPPING THE WALK BEHIND TRACTOR

The walk behind tractor can be stopped by using the clutch lever as follows:

STOPPING THE WALK BEHIND TRACTOR IN CASE OF PLOWING

1. Shift the accelerator lever to decrease the engine speed.
2. Pull the main clutch for model single clutch or the right clutch for model dual clutch to the “**DISENGAGED**” position.

STOPPING THE WALK BEHIND TRACTOR IN CASE OF TRAVELING OR HITCHING

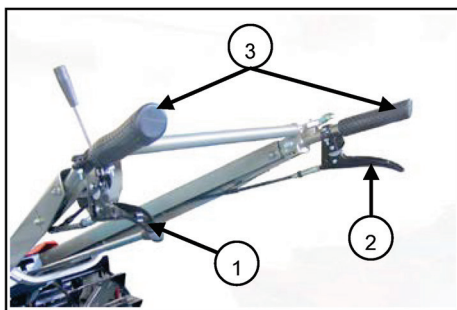
1. Shift the accelerator lever to decrease the engine speed.
2. Pull the main clutch for model single clutch or the right clutch for model dual clutch lever to the “**DISENGAGED**” position.
3. The walk behind tractor may not stop yet. Then slowly pull the main clutch lever for model single clutch or the right clutch lever for model dual clutch to the “**STOP**” position. The walk behind tractor can stop as desired.

LEFT-RIGHT STEERING CONTROL

The movement direction of the walk behind tractor can be controlled by using the steering lever, consisting of 2 levers:

“Left steering lever” When pressing the left steering lever, the left wheel will stop, prompting the walk behind tractor to turn left.

“Right steering lever” When pressing the right steering lever, the right wheel will stop, prompting the walk behind tractor to turn right.



(1) Left steering lever
(2) Right steering lever
(3) Steering handle

To steer the walk behind tractor, firmly press the steering lever and use the steering handle only.

⚠ CAUTION

- Do not squeeze the steering lever while the walk behind tractor is on a steep slope, hitching with a trailer, traveling at high speed, or on a curvy road. Otherwise, the walk behind tractor may make a sudden turn and topple over.

■ NOTE

- The steering control of the walk behind tractor model single clutch and dual clutch shares the same operation.

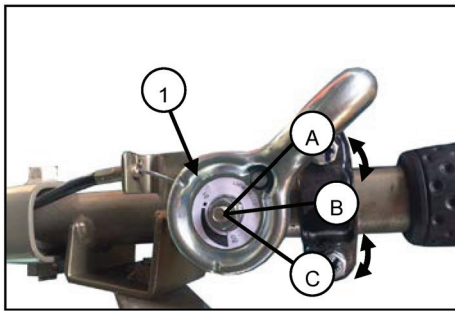
ACCELERATING THE ENGINE SPEED

The engine speed can be accelerated by using the accelerator lever, consisting of 3 speeds:

“STOP” position is used for stopping the engine.

“DECELERATE” position is used for decelerating the engine speed with low speed.

“ACCELERATE” position is used for accelerating the engine speed with high speed.



(1) Accelerator lever
(A) “STOP” position
(B) “DECELERATE” position
(C) “ACCELERATE” position

To stop the engine, shift the accelerator lever to the **“STOP”** position, and before shifting the accelerator lever to the **“STOP”** position, follow these steps:

1. Place the clutch lever to the **“DISENGAGED”** position to stop the walk behind tractor.
2. Place the gear lever to the **“NEUTRAL”** position.

STARTING THE WALK BEHIND TRACTOR ENGINE

Before starting the engine, check the main clutch at the **“DISENGAGED”** or **“STOP”** position only and the gear lever at the **“NEUTRAL”** position. Then accelerate the engine speed to the **“START”** position and start the engine. To start the engine, follow the instructions in “Operation and Maintenance Manual for KUBOTA diesel engine”

OPERATING THE WALK BEHIND TRACTOR IN DIFFERENT FIELD CONDITIONS

To operate the walk behind tractor in different field conditions, follow these instructions:

1. In normal paddy fields, when the walk behind tractor is attached to the implements, such as mouldboard plow, disc plow or harrow with the ski, it is possible to use the 2nd gear, provided that the engine runs smoothly without hesitation or a sharp drop of engine speed.
2. In middle mire-soil paddy fields, when the walk behind tractor is attached to the implements mentioned above and the engine is found of hesitation or a sharp drop in engine speed, stop the walk behind tractor and shift into the 1st gear to maximize the engine power, allowing the vehicle to drive through the mire. The 2nd gear can then be operated.
3. In much mire-soil fields, if the operation cannot follow the instructions in No. 2 above, install additional walk behind tractor wheel with 80 cm. diameter (general tractor's wheel). Pair it with each side of the wheel and continue operating the walk behind tractor.

CHECKING BEFORE OPERATION

■ IMPORTANT

- **Carefully check parts of the walk behind tractor as the instructions in this manual to obtain the best use of the machine.**
1. Check the completion and condition of the walk behind tractor. If any part is found worn out or damaged from operation, repair or replace a new one before operating the walk behind tractor.
 2. Always check and tighten every pin and nut.
 3. Control cables which are main clutch cable, steering cable and accelerator cable must be lubricated with oil to smoothen the control.
 4. Check the transmission case leaks at each part. If found, repair it.
 5. Check the belt tension as required. (For adjusting the belt tension, refer to “Adjusting the belt tension” on page 33.)
 6. For checking before operation of the engine attached with the walk behind tractor, refer to “Operation and Maintenance for KUBOTA diesel engine”.

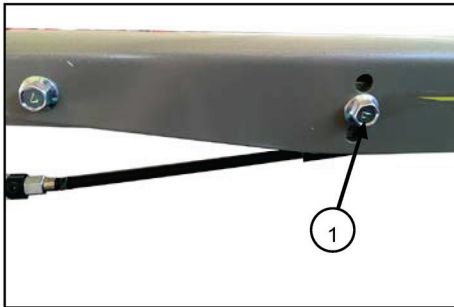
ADJUSTING PARTS OF THE WALK BEHIND TRACTOR

CAUTION

- To prevent accidents, always stop the engine before making any adjustments to the walk behind tractor.

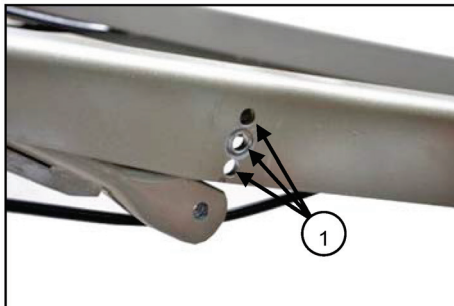
ADJUSTING THE HANDLE OF THE WALK BEHIND TRACTOR

1. Loosen the handle securing nut from both sides.



(1) Handle securing nut

2. Choose the desired position of the handle adjusting hole. Then re-tighten the nut.



(1) Handle adjusting hole

ADJUSTING THE HANDLE STAY OF THE WALK BEHIND TRACTOR

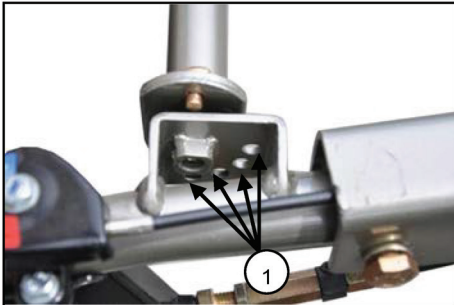
The handle stay can be adjusted to 4 positions as follows:

1. Loosen the handle stay securing lock nut until the spindle attached with the locking plate is out of the hole.



(1) Handle stay securing lock nut

2. Turn to choose one of four positions of the handle stay adjusting hole. Adjust the spindle into the locking plate hole and re-tighten it.



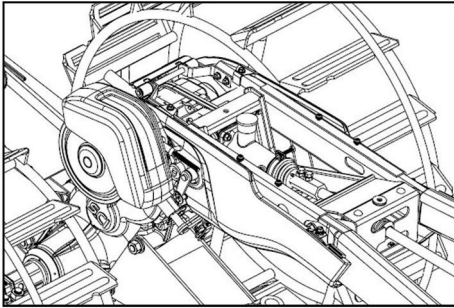
(1) Handle stay adjusting hole

ADJUSTING THE MAIN CLUTCH LEVER

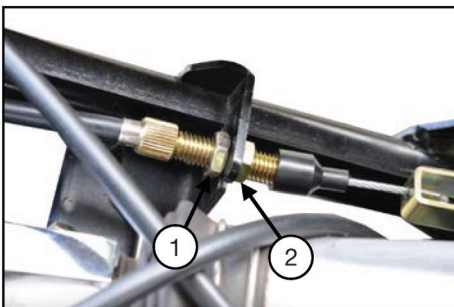
During operation, if the main clutch lever is pulled to the **“DISENGAGED”** position yet difficult to be engaged, loud noise is observed, or the clutch is released to the **“ENGAGED”** position yet the walk behind tractor fails to move, re-adjust the clutch free play as follows:

ADJUSTING THE MAIN CLUTCH LEVER FOR THE WALK BEHIND TRACTOR MODEL SINGLE CLUTCH

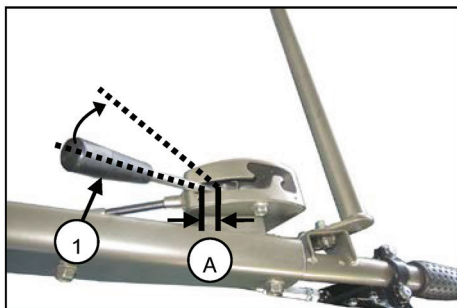
1. Open the cover.



2. Loosen the clutch cable lock nut and turn the main clutch cable adjuster. Then recheck the clutch free play by engaging the main clutch to the **“ENGAGED”** position and slowly pulling it to the **“DISENGAGED”** position until slight resistance is felt.

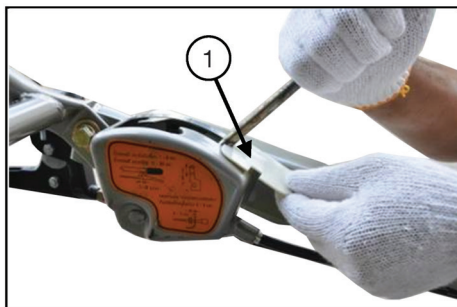


(1) Clutch cable lock nut
(2) Clutch cable adjuster

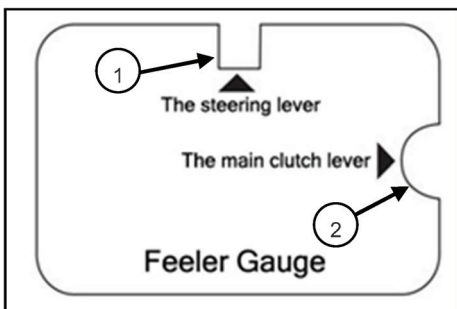


(1) Main clutch lever
(A) Free play

3. Adjust the clutch free play as required or use the feeler gauge while adjusting. Tighten the clutch cable lock nut.



(1) Feeler gauge



(1) For adjusting the steering lever
(2) For adjusting the main clutch lever

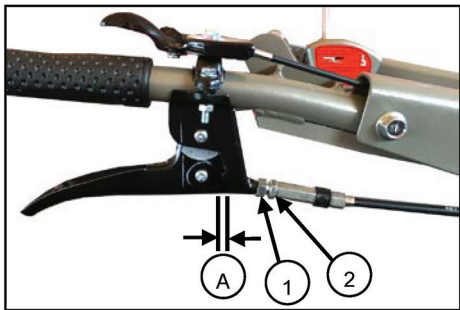
Main clutch lever free play	15-20 millimeters
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4. Test the operation of the clutch system by running the walk behind tractor.

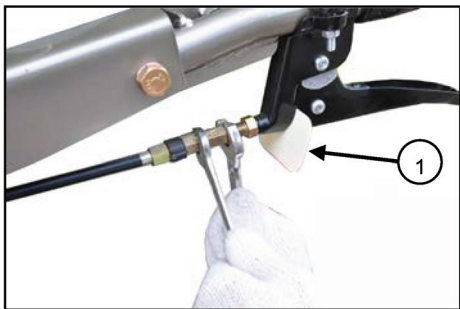
ADJUSTING THE LEFT-RIGHT STEERING LEVER

During operation, if the steering lever is pressed yet the walk behind tractor is not turned, the steering lever has no or too much free play, re-adjust the steering lever free play as follows:

- 1. Loosen the adjuster lock nut.
- 2. Rotate the adjuster and check the steering lever free play, starting from pressing the steering lever until slight resistance is felt.
- 3. Measure and adjust the left-right steering lever free play as required or use the feeler gauge while adjusting.
- 4. Tighten the adjuster lock nut.



(1) Lock nut
(2) Adjuster
(A) Free play at 1-2 millimeters



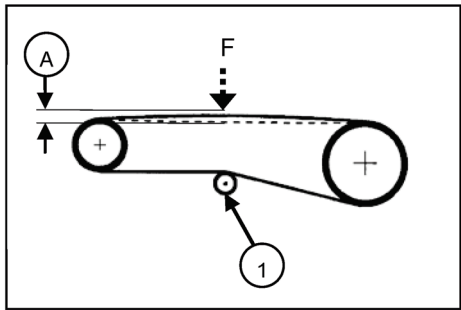
(1) Feeler gauge

Steering lever free play	1-2 millimeters
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ADJUSTING THE BELT TENSION

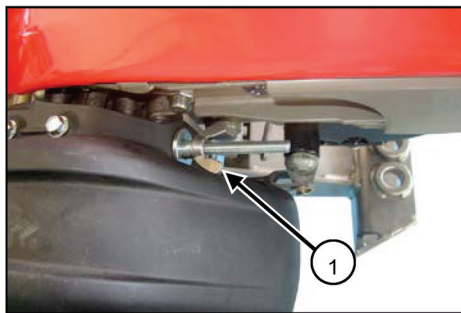
After hours of operation, the belt may become loosened, reducing the plowing or dragging power or causing belt slipping out of the pulley. It is, therefore, necessary to always adjust the belt tension as follows:

- 1. Check the belt tension by pressing the finger on the middle of the front belt.



(1) Engine pulley
(A) Belt tension at
30-40 millimeters

- 2. If the deflection is higher or lower than the requirement, tighten the belt tension adjusting wing bolt to lift up the pulley until the belt is stretched. Measure and adjust the belt tension as required.

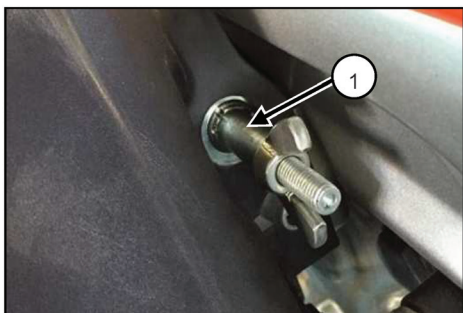


(1) Belt tension adjusting wing bolt

Belt tension	30-40 millimeters
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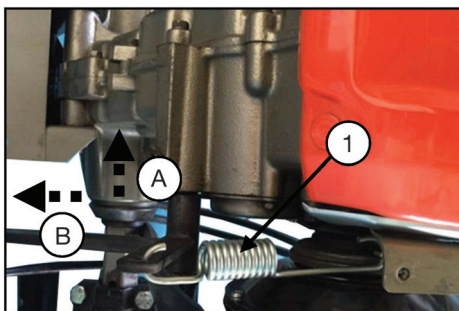
CAUTION

- When assembling the wear sleeve on the belt adjustment lever, make sure to assemble the wear sleeve before the belt cover.



(1) Wear sleeve

- When assembling the tensioner spring, face the end of the spring toward the gear case side, and face the spring coil side toward the front of the walk behind tractor.



(1) Spring

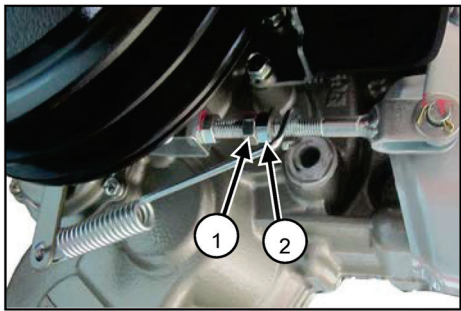
(A) Spring coil side

(B) Front side of walk behind tractor

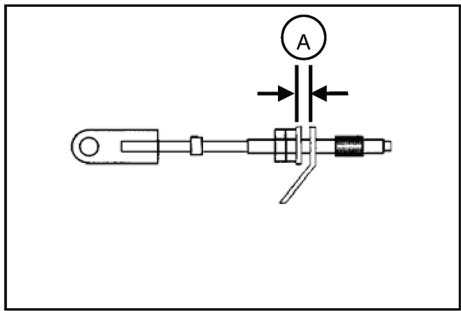
ADJUSTING THE BRAKE FREE PLAY

When the main clutch lever is pulled to the “**STOP**” position yet the walk behind tractor fails to stop or when the main clutch lever is pulled to the “**DISENGAGED**” position yet the brake system is operating, adjust it as follows:

- 1. Pull the main clutch lever to the “**DISENGAGED**” position.
- 2. Loosen the brake adjustment lock nut
- 3. Adjust the brake free play by adjusting the brake adjustment lock nut as required and tighten it.



(1) Brake adjustment lock nut
(2) Brake adjustment nut



(A) Brake free play at
3-5 millimeters

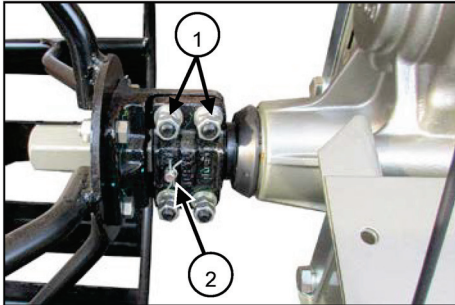
- 4. Test the operation of the brake system by placing the gear lever to the “**NEUTRAL**” position and pulling the main clutch lever to the “**STOP**” position. Then try moving the walk behind tractor backwards or forwards. If the walk behind tractor does not move it means the brake system functions properly.

Free play	3-5 millimeters
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ADJUSTING THE WHEEL SPACING AND CHANGING THE WHEEL

The wheel spacing can be adjusted at 3 levels as proper and desired. The steps of adjusting the wheel spacing are as follows:

1. Park the walk behind tractor on the flat ground, loosen four wheel hub lock nuts of a side.



(1) Wheel hub lock nut

(2) Snap pin and wheel hub pin

2. Adjust the wheel spacing or replace the wheel as desired.
3. Place the wheel hub pin and snap pin back to their positions and tighten four wheel hub lock nuts.
4. For the other wheel, follow the same procedures 1-3 as mentioned above.

■ IMPORTANT

- **Before the operation, make sure the snap pin, wheel hub pin, and wheel hub lock nuts are firmly tightened. To maintain the tractor balance, always adjust the track width so that both left and right sides have equal spacing.**

ADJUSTING THE WALK BEHIND TRACTOR BALANCE

When operating the walk behind tractor with other implements, check if the walk behind tractor is properly counterweighted as follows:

If the front end is too heavy,

Check the engine position. If practical, move the engine to the back of the walk behind tractor until it is lighter. Then adjust the belt tension as required.

If the rear end is too heavy,

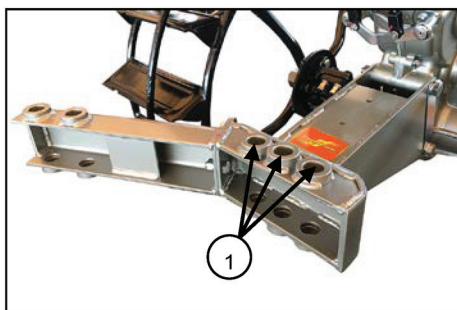
Check the engine position. If practical, move the engine to the front of the walk behind tractor until it is lighter. Then adjust the belt tension as required.

INSTALLING AND OPERATING THE IMPLEMENTS

CONNECTING THE IMPLEMENTS

(For model NC Plus X only)

Insert the pin hole of the implement into the connecting hole of the walk behind tractor and place the pin to lock it.

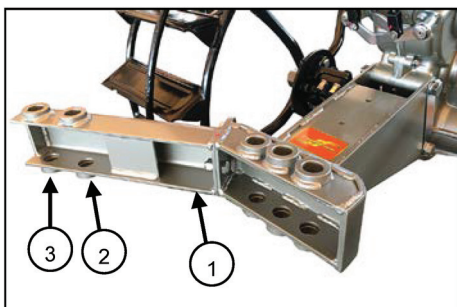


(1) Connecting hole

CONNECTING THE IMPLEMENTS USING THE EXTENSION HITCH

(For model NC Plus X only)

It is used for plowing and leveling the field, suitable for operating with the disc plow and mouldboard plow. The plow will be at the wheel for plowing the soil left from the right wheel trail which allow the walk behind tractor to run after to the right side.



(1) Extension hitch

(2) Hole for disc plow

(3) Hole for mouldboard plow

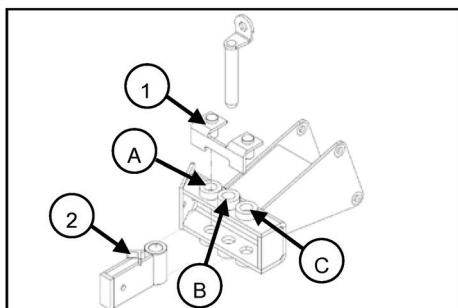
STOPPER PLATE

In case other non-KUBOTA implements, such as mouldboard plow and harrow, etc. are used, install the stopper plate to prevent the implement from hitting the wheels.

Installing the stopper plate to the hitch bar

Install the stopper plate as follows:

- To install the implement to the left hitch-pin hole (A), install stopper plate at the left.
- To install the implement to the center hitch-pin hole (B), install the stopper plate at the center.
- To install the implement to the right hitch-pin hole (C), install the stopper plate at the right.



- (1) Stopper plate
- (2) Implement arm
- (A) Left position
- (B) Center position
- (C) Right position

SERVICE INTERVALS AND LUBRICANTS

SERVICE INTERVALS

Number	Maintenance items		Service hour		See page
			Every time before operation	Every	
1	Transmission oil	Check	✓	-	42
		Change*	-	500 hour	
2	Lubricating parts	Lubricate	Every time before operation		43
3	Clutch lever	Check/ Adjust			30
4	Left-right steering lever	Check/ Adjust			32
5	Belt tension	Check/ Adjust			33
6	Brake free play	Check/ Adjust			35

* At the first 50 operating hours, and every 500 operating hours afterward.

LUBRICANTS

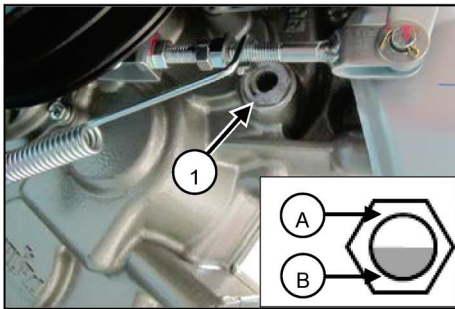
	Locations	Capacity	Lubricants
1	Transmission oil	5.5 - 6 liters	SAE EP90 API GL-4
2	Lubricating parts	-	TRA CHANG SAE 30 or 40 engine oil
3	Applying grease	-	Grease number 2

PERIODIC SERVICE

CHECKING AND CHANGING TRANSMISSION OIL

CHECK THE TRANSMISSION OIL LEVEL

1. Park the walk behind tractor on the flat ground horizontally and check the transmission oil level at the level check port. If the transmission oil is lower than the requirement, fill up the oil until it reaches the requirement.

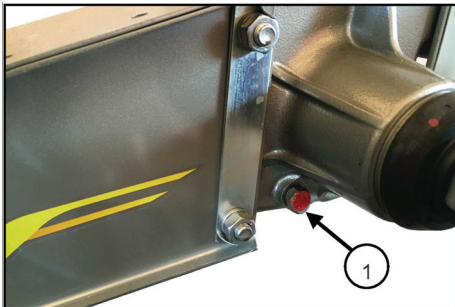


(1) Level check port
(A) Highest level
(B) Lowest level

CHANGING THE TRANSMISSION OIL

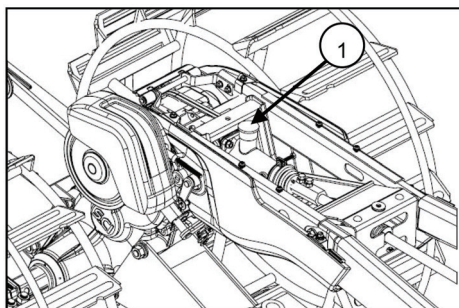
In case purchasing or repair the whole machine, change the transmission oil when operating for first 50 hours and every 500 hour next time.

1. Remove the oil level check plug on the left side under the transmission case to drain all transmission oil from the transmission case.



(1) Oil level check plug

2. Re-tighten the oil level check plug. Then fill up the oil through the transmission oil inlet until it reaches the requirement. Re-tighten the transmission oil filter plug.



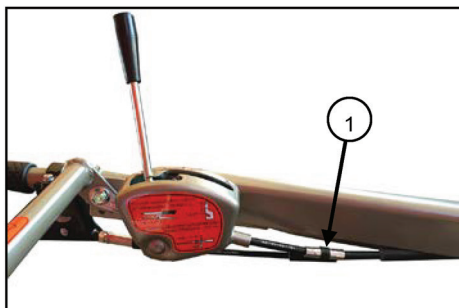
(1) Transmission oil filter plug

3. Check the transmission oil level as required before operating the walk behind tractor.

LUBRICATING PARTS

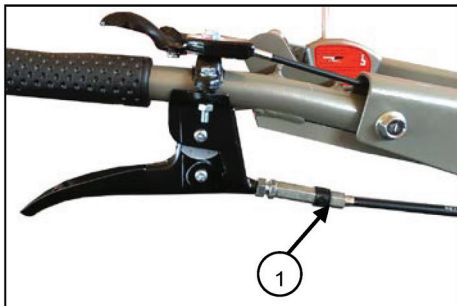
The moving parts and friction-bearing joints require regular lubrication with TRA CHANG SAE 30 or 40 engine oil as follows:

1. Lubricate the main clutch lever cable.



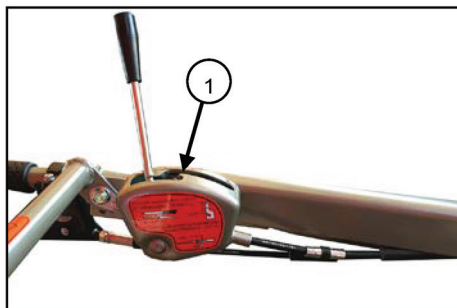
(1) Lubricating parts

2. Lubricate the left-right steering lever.



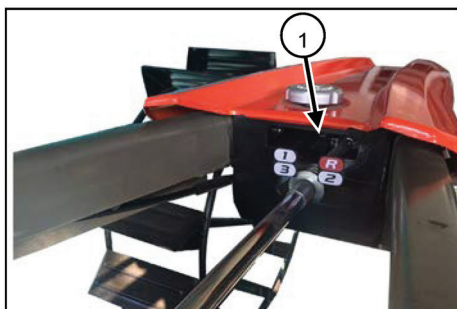
(1) Lubricating parts

3. Apply grease to the main clutch lever.



(1) Grease apply locations

4. Apply grease to the gear shift lever.



(1) Grease apply locations

5. Lubricate the moving or rotating parts of the walk behind tractor.

TROUBLESHOOTING

Trouble	Cause and Countermeasure
1. Loud noise while engaging the gear	- The gear is shifted while the walk behind tractor does not stop still, for model single clutch. Or the left clutch lever is not firmly pressed while shifting the gear, for model dual clutch. - The clutch free play is adjusted incorrectly. Re-adjust it. - The transmission oil is too low. Check and refill the oil as required. - Wrong number of transmission oil is filled up or the oil is not changed following the service intervals. Follow the instructions in the manual. - The bearing or gear is damaged during operation.
2. The walk behind tractor does not move or has no power.	- Check the gear shift lever not to be at the "NEUTRAL" position. - Check the clutch lever position at the "ENGAGED" position. - The clutch plate is worn out or dirty with oil. Replace it. - The belt between the engine and the walk behind tractor is loosened. Re-adjust it. - Accelerate the engine speed too low. Accelerate it properly. - The clutch free play is adjusted incorrectly. Re-adjust it.
3. The walk behind tractor cannot be stopped.	- The clutch free play is adjusted incorrectly. Re-adjust it. - The brake plate is worn out. Replace it.
4. The wheel shaft seal is leaked.	- Sand enters the wheel seal. Replace it. - It was worn out according to the operating condition. Replace it.
5. The oil leaks out of the seal of the gear shift lever cover.	- The wrong number of transmission oil is filled up or it is much more than the requirement. Follow the instructions in the manual.

Trouble	Cause and Countermeasure
6. The walk behind tractor turns itself.	- The steering lever free play is adjusted incorrectly. Re-adjust it. - The implement is adjusted incorrectly. Re-adjust it. - The steering gear is damaged.

■ **NOTE**

- If the problem cannot be solved after following the preliminary instructions, please contact **KUBOTA** dealers or technical service center for service.

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